

INTERIOR ANGLES OF A REGULAR POLYGON



LESSON INTRODUCTION

MA.8.GR.1.6 Develop and use formulas for the sums of the interior angles of regular polygons by decomposing them into triangles.

LEARNING TARGETS

- I can use formulas to find the sum of the measures of the interior angles of a regular n -gon.
- I can determine the measure of an interior angle in a regular n -gon.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How do I find the sum of the measures of the interior angles of a regular n -gon?
- How can I find the measure of an interior angle in a regular n -gon?

LESSON NARRATIVE

Students continue their work with interior angle measures of various regular polygons. Students will collaborate in pairs on an activity focused on the interior angles of regular polygons, including the introduction of the sum of the interior angles of a n -gon. Students will get the opportunity to engage in guided practice with interior angle sums and individual angle measures before independently practicing their understanding of sums and angle measures of regular n -gons.

COMPONENT SUMMARY

13.6.1 WARM-UP (~5 MIN)

Create a tiling

13.6.3 GUIDED PRACTICE (15 MIN)

Find interior angle sums and individual angle measures

13.6.5 WRAP-UP (~5 MIN)

Write a note after analyzing student work to identify an error

13.6.2 COLLABORATION (15 MIN)

Sum of interior angles of a n -gon

13.6.4 YOUR TURN! (10 MIN)

Practice finding interior angle measures

made

RESOURCES

GLOSSARY

Key Terms:

- Sum of the Interior Angles

Additional Terms:

- N/A

MATERIALS

- (Optional) Markers or colored pencils

ADDITIONAL PREPARATION

- Consider grouping students based on formative assessment data from Lesson 5.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recall the formula for the sum of interior angles from Lesson 5.
- Students may forget to subtract 2 from the number of sides prior to multiplying by 180.
- Students may make calculation errors when determining the sum of the interior angles.

THINGS TO CONSIDER

- Students learned the sum of the interior angles formula in Lesson 5. This lesson is an extension of that to regular n -gon's. Consider using some of this time to support students who struggled with the concept in the prior lesson.

LITERACY AND LANGUAGE ROUTINES

Error Analysis

13.6.5 Wrap-Up provides students with an opportunity to complete an error analysis. Students identify and correct errors in a sample of student work based on their understanding of finding the measure of one interior angle of a regular polygon. After correcting the errors, students will be less likely to make similar errors themselves in the future.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR1.1 Participate in Effortful Learning

13.6.2 Collaboration provides students with an opportunity to collaborate with their peers on finding the interior angles of regular polygons. During this activity, students analyze the questions, support each other, and persist to solve the questions together.

DIFFERENTIATE INSTRUCTION

Consider using formative assessment data from Lesson 5 to determine the groupings for 13.6.2 Collaboration. Since 13.6.2 Collaboration builds on what students learned in Lesson 5, this can create opportunities for struggling students to get support from their peers as they explore regular n -gons.

13.6.1 WARM-UP: TILES AND REGULAR POLYGONS

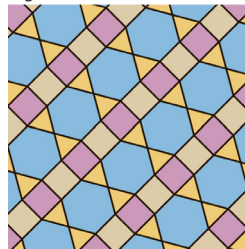
TEACHER MOVES

For this activity, it is important that students can identify the different polygons in question 1. In question 2, the tiling that students create isn't limited to the shapes they've experienced. Instead, students should be encouraged to be creative, using different geometric shapes, coloring their tessellations with colored pencils or markers.



13.6.1 Warm-Up: Tiles and Regular Polygons

A tiling consists of covering a surface using copies of shapes, called tiles, without overlapping or leaving gaps. A picture of a tiling is shown.

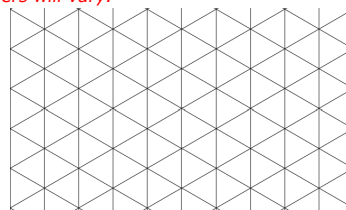


- Trace over and name each of the different regular polygons you notice in the picture.

equilateral triangle, square, hexagon

- Use the grid to create a tiling.

Answers will vary.



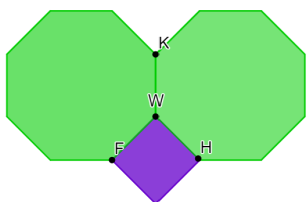


13.6.2 Collaboration: Interior Angles of Regular Polygons



The **sum of the interior angles** of an n -gon can be found using the formula $180(n - 2)$, where n is the number of sides of the polygon.

1. Jerrica plans to use a design similar to the one shown to tile her bathroom floor.



- a. Work with your partner to find the interior angle sums of each polygon included in the tile design. Show your work in the space provided.

Regular Octagon	Square
$(8 - 2) \cdot 180 = 1080^\circ$	$(4 - 2) \cdot 180 = 360^\circ$

- b. Work with your partner to determine the measure of each angle.

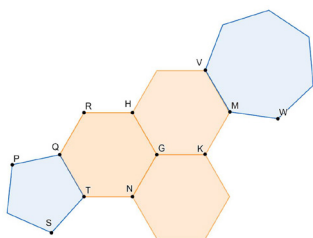
Measure of $\angle FWH = 90^\circ$

Measure of $\angle FWK = 135^\circ$

Measure of $\angle HWK = 135^\circ$

- c. What type of angles are $\angle FWH$ and $\angle HWK$?
- ☒ adjacent
 - ☐ complementary
 - ☐ supplementary
 - ☐ vertical

2. A diagram of regular polygons is shown.



Find the measure of each sum.

a. $m\angle STQ + m\angle NTQ = 228^\circ$

b. $m\angle NGH + m\angle HGK = 240^\circ$

c. $m\angle PQT + m\angle QRH + m\angle VMW = 356.571^\circ$

13.6.2 COLLABORATION: INTERIOR ANGLES OF REGULAR POLYGONS

DIFFERENTIATE INSTRUCTION

Consider using formative assessment data from Lesson 5 to determine the groupings for 13.6.2 Collaboration. Since this builds on what students learned in Lesson 5, it provides opportunities for struggling students to get support from their peers as they explore regular n -gons.

DIFFERENTIATE INSTRUCTION

Consider having students highlight the angles using markers or colored pencils to see the angle they are working with.

ENRICHMENT

- How can you determine the measure of the angles being added?
- How can you use the sum of the angles for each polygon to determine each angle?

13.6.3 GUIDED PRACTICE: INTERIOR ANGLE SUMS AND INDIVIDUAL ANGLE MEASURES

DIFFERENTIATE INSTRUCTION

Use observational data from the 13.6.2 Collaboration to determine how much guidance is needed to bring together key ideas in this component. If necessary, consider modeling question 1 to address any common misconceptions that arose in the Collaboration.

ENRICHMENT

- How are we able to determine the sum of the measures of the interior angles of this polygon without an image?
- How can you use the number of sides given to determine the measure of one interior angle of a regular polygon?

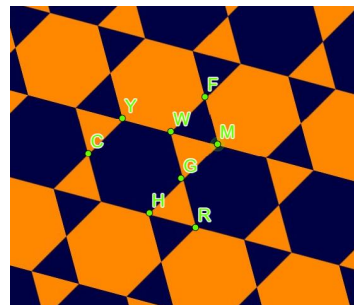
TEACHER MOVES

Consider providing additional instructions to use a specific name for the polygons when identifying. For example, students may say “triangle,” but it should be “equilateral triangles.”



13.6.3 Guided Practice: Interior Angle Sums and Individual Angle Measures

1. A regular polygon has 14 sides.
 - a. What is the sum of the measures of the interior angles of this polygon?
2160°
 - b. What is the measure of one interior angle of this polygon?
154.286°
2. The tile pattern uses regular polygons.



- a. What regular polygons are in the tile pattern?
equilateral triangles, hexagons

- b. Find the measure of each.

$$m\angle FWY = \underline{120^\circ}$$

$$m\angle HGW + m\angle HGR = \underline{180^\circ}$$

$$m\angle MGR + m\angle GRH = \underline{180^\circ}$$

3. The Pentagon is the headquarters building of the U.S. Department of Defense in Washington, D.C. Each of the 5 outside walls measures 921 feet, making it a regular pentagon.



Curtis plans to build a scale model of the Pentagon. What is the measure of the angle where any two outside walls join?

108°

13.6.3 GUIDED PRACTICE: INTERIOR ANGLE SUMS AND INDIVIDUAL ANGLE MEASURES (CONTINUED)

TEACHER MOVES

Students may struggle when they transition to question 3b. They may not quickly realize that they have what they need to solve this question. If needed, scaffold support to help students understand they have all the information they need.

ENRICHMENT

- What information provided will be helpful in finding the measure of the angle where any two outside walls join?
- What information is not needed?

13.6.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider using this time to help students who are struggling and may need additional support. This is an opportunity to provide a small group with additional support.



13.6.4 Your Turn!

1. What are the measures of each interior angle of a regular dodecagon, a polygon with 12 sides?
 150°
2. How many degrees greater is the measure of one interior angle of a regular nonagon, a polygon with 9 sides, than the measure of one interior angle of a regular octagon, a polygon with 8 sides?
 5°



13.6.5 Wrap-Up: Error Analysis

1. Cathy is finding the measure of one interior angle of a regular pentagon. Her thinking and work are shown.

I can draw the diagonals from one vertex to form 4 triangles.

The sum of the measures of the interior angles is $4 \times 180^\circ = 720^\circ$.

The pentagon has five equal angles, so each angle measures $\frac{720^\circ}{5} = 144^\circ$.

Her solution is incorrect. Write a note to Cathy to explain the mistake she made. Then, provide her with the correct steps.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

The mistake was in finding the number of triangles inside a regular pentagon. You can only form 3 triangles in a pentagon.

The sum of the measures of the interior angles is $3 \times 180^\circ = 540^\circ$ using the 3 triangles, instead of 4.

The pentagon has five equal angles, so each angle measures $\frac{540^\circ}{5} = 108^\circ$.

13.6.5 WRAP-UP: ERROR ANALYSIS

DIFFERENTIATE INSTRUCTION

Provide sentence stems to help students compose their notes. For example:

- Cathy made a mistake when she _____ because _____.
- Cathy should have _____ instead of _____.
- The next time Cathy sees _____, she should _____.